

KENISYA SARAVENEN

Biomedical Sciences Undergraduate | University of York |

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Biomedical Sciences undergraduate at the University of York with a First-class average in Year 1 and academic/practical experience across genetics, molecular biology, physiology and microbiology. Experience includes R-based biological data analysis, antimicrobial susceptibility testing, PCR, agarose gel electrophoresis, scientific report writing and exposure to pharmaceutical and public-health settings. Seeking a 2027-28 Year in Industry placement across biomedical research, pharmaceuticals, biotechnology, diagnostics or genomics.

EDUCATION

University of York | BSc Biomedical Sciences | 2025 - Present

- First-class (1st, 1:1) average in Year 1 | Expected graduation: 2029
- **Relevant Coursework:** Physiology, Genetics, Molecular Biology, Cell biology, Neuroscience, Immunology, Mechanisms of disease.
- *Recombinant DNA technology · DNA sequencing methods · Cell signalling pathways · Stem cell differentiation · Cellular immunology mechanisms · Enzyme kinetics and inhibition · Hepatic detoxification pathways*

Monk's Walk School 6th Form | A Levels | 2023 - 2025

- Biology A | Chemistry B | Mathematics B

Relevant GCSEs: Biology 9 | Chemistry 9 | Geography 9 | Mathematics 8 | Physics 7

SKILLS

- **Laboratory & Clinical:** Aseptic technique, microbial culturing, disk diffusion assays, gel electrophoresis, PCR, spectrophotometry, PowerLab acquisition (ECG, EMG, spirometry), GLP
- **Biostatistics & R:** Data wrangling (tidyverse, readxl), statistical modeling (linear models, one-way ANOVA, emmeans, post-hoc pairwise comparisons), custom data visualization
- **Scientific Communication:** Evidence-based scientific report writing, academic poster presentations, project coordination, and cross-functional teamwork
- **Core Competencies:** Analytical problem solving, adaptable under pressure, strict attention to detail, and time management
- **Software:** Microsoft 365 (Excel, Word, PowerPoint) and Google Workspace (Sheets, Docs, Slides)

RELEVANT EXPERIENCE

Public Health England | Biomedical Science Internship | July 2025

- Gained practical exposure to biomedical science workflows within a public-health laboratory setting.
- Supported organisation and standardisation of Streptococcus pneumoniae agar plate workflows.
- Supported laboratory work relating to influenza serotype identification and PCR-based analysis of genetic sequences associated with Streptococcus pneumoniae.
- Developed understanding of laboratory practice, research workflows and the role of scientific evidence in public health.

Sunway University - Qualitas Medical Group, Malaysia | Clinical Shadowing | August 2024

- Shadowed a general practitioner and observed patient consultations, clinical workflows and the application of biomedical knowledge in healthcare.
- Observed professional communication and patient-facing practice, developing awareness of confidentiality, ethics and professional conduct in clinical settings.
- Gained insight into the relationship between scientific knowledge, clinical practice and patient care.

Roche Pharmaceuticals | Data & Analytical Sciences Workshop | July 2024

- Completed an industry workshop focused on data and analytical sciences within the pharmaceutical sector, gaining insight into how analytical approaches support biomedical research and drug development.
- Developed an understanding of the drug development pipeline, including key stages from early drug discovery and preclinical development through clinical trial Phases I-III, regulatory approval and launching medicines into the market.
- Gained insight into the application of data-driven and analytical approaches within a global pharmaceutical environment, with consideration of their role in informing research, development and decision-making.

ADDITIONAL EXPERIENCE

Private Tutor | Tutor Hunt | 2026 - Present

- Tutor Mathematics, Biology and Chemistry to students across a broad age range.
- Plan and deliver structured lessons tailored to individual learning needs, adapting explanations and questioning to build understanding.
- Manage multiple learners independently, demonstrating organisation, reliability, communication and time management.

British Heart Foundation | Volunteer | June 2024 - January 2025

- Supported customer service, till operations, stock organisation and day to day shop operations.
- Communicated confidently with customers of varied ages and backgrounds while working collaboratively with staff and volunteers.

RESEARCH & ACADEMIC WORKS

Investigation of the Antibacterial Effects of Trimethoprim and Sulfamethoxazole Against *E. coli* | University of York | 2025

- Investigated environmental and microbiological indicators of water quality through phosphate/nitrate analysis, total viable counts, Nutrient agar culture and *E. coli* identification using EMB agar.
- Applied colony PCR and agarose gel electrophoresis to investigate beta-lactamase/ampicillin resistance related samples and interpret experimental results.
- Investigated the antibacterial activity of trimethoprim (TMP), sulfamethoxazole (SMX) and combined TMP+SMX treatments against *E. coli* using disk diffusion assays at 10, 100 and 500 µg/mL, incorporating positive and negative controls and four experimental replicates.
- Analysed experimental data in **R 4.4.1** using **tidyverse**, **readxl** and **emmeans**, applying linear modelling, one-way ANOVA and post-hoc pairwise comparisons; identified a significant treatment effect (**p < 0.001**, **R² = 0.9927**).
- Authored an **article-style scientific report** critically evaluating concentration-dependent antibacterial activity, combination treatment effects, experimental limitations and implications for antimicrobial resistance.

Health & Longevity Group Project | University of York | 2026

- Led group scientific direction and project logistics for a health support application concept, helping guide development of a meal scanning feature designed to return evidence-based health and cognitive information on identified ingredients.
- Researched and synthesised scientific evidence on ingredient-specific health and brain-related effects to inform application content and scientific rationale.
- Coordinated group contributions and presented the project, combining scientific communication, project leadership and evidence-based decision-making.

Cancer Informatics & Bioinformatics Strand | University of York | Semester 2, 2026 - 27

- Will analyse large-scale DNA/RNA sequencing and clinical datasets in R, developing experience in genomic profiling, statistical analysis, clinical data integration and machine learning-based cancer classification.
- Will investigate molecular features that distinguish cancer types and explore clinically relevant applications of sequencing data, including cancers of unknown primary origin.

SCIENCE COMMUNICATION

- Produced a feature exploring the **gut-brain axis**, examining evidence linking the gut microbiome and diet with cognitive health and Alzheimer's disease risk, while communicating the multifactorial nature of Alzheimer's disease and the limitations of microbiome-based prevention.
- Wrote a public facing blog exploring **electrical signalling in fungi**, explaining how signals propagate through interconnected hyphae and may coordinate responses to environmental changes and threats, while translating emerging research into accessible scientific content.
- Developed an accessible explainer on **PCR**, outlining denaturation, annealing and elongation, the roles of primers, DNA polymerase and nucleotides, and the importance of DNA amplification in molecular biology and biomedical research.

LEADERSHIP & ACTIVITIES

Head of Wellbeing | College Volleyball Club | 2026 - 2027

- Committee responsibility supporting member wellbeing, communication and an inclusive club environment.

Social Media Officer | York University - James College Pool | 2026 - 2027

- Manage social-media content and contribute to communication and engagement for the college pool society.

ADDITIONAL INFORMATION

Languages: English (Fluent), Korean (Intermediate Reading & Writing), Malay (Intermediate Speaking)

Interests: Piano (Grade 5), University sport and societies, Karate, Taekwondo and Krav maga, Bronze Duke of Edinburgh Award